Research Methodology & IPR

Debesh Choudhury

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Research $\xleftrightarrow{\text{They are interconnected}}$ Truth

A Course for M.Tech. in Biomedical Instrumentation, Computer Science & Engineering, and Optics & Optoelectronics

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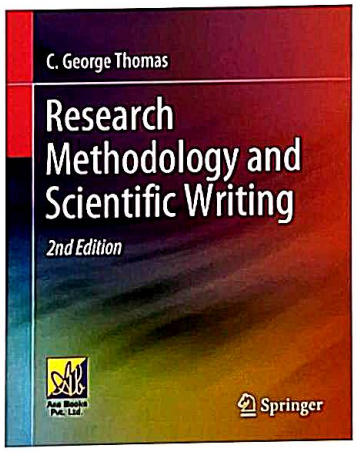
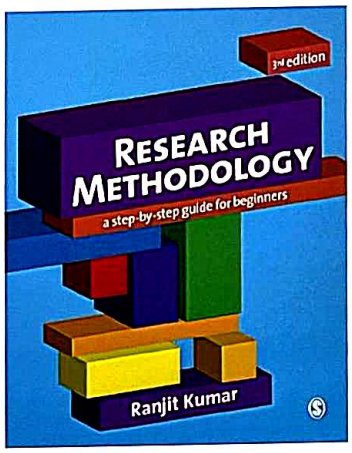
1. Overview
2. Research
3. Sources
4. Traits
5. Errors
6. Scope
7. History
8. Art, Literature
9. Science
10. People

Reference books





Meaning of research

Research is used to denote

- defining, redefining, and solving problems;
- observing facts and their interpretation;
- formulation of hypotheses and their testing through experiments;
- revision of existing theories and laws;
- practical application of information already generated.

Research has a PURPOSE

To help the organization

In most scientific disciplines, research is used as a tool

- to understand and solve problems for the benefit of people;
- to manage defence for chalking out better strategies;
- to use it in solving and settling issues related to profession.

To help the community

To help the society

Motives of research

There are plenty of motivation for research

- Excitement of discovery
- Pursuit of prestige
- In depth knowledge on a subject
- Service to society
- Need for publications
- Obtaining a higher degree
- Better management
- A tool in defence
- Research in everyday life (for professionals)

Newton, Kepler, Galileo, Franklin, Faraday, ...

Maxwell, Volta, Graham Bell, J. C. Bose, ...

Oppenheimer, Einstein, S. N. Bose, Tesla, ...

professors, me, you, professionals, ...

Benefits of conducting research

The society enjoys the following benefits

- Expanding frontiers of knowledge — scientific disciplines expands
- New inventions and discoveries — main objective of research
- Solving problems affecting the society — solve societal problems
- Increasing efficiency and reducing costs — improve production
- Research strives to make life easier — improve lives of the people
- Luxury and comfort — better living standard
- Infotainment — computers, Internet, smartphones, etc
- Economic growth — societal growth

Sources of a research problem

Primary sources of a research problem

- Existing Literature — reviews of past research works
- Personal Experience and Professional Practice — self experience and supervisor's advice
- Social, Cultural, and Professional Contexts — contexts pave down to new research paths
- Sophisticated Technologies and Innovations — new tech opens up new research
- Observations of Everyday Life — life inspires new research

Characteristics of good research

Good research is characterized by certain attributes

- Research is based on the work of others — ensures good literature survey
- Research is a blend of logic and imagination — logical imagination
- Research tries to identify and avoid bias — impartial and not biased
- Repeatability — if we repeat, we get the same results
- Research is systematic — structured and systematic
- Research generates new questions — ask questions and provide answers
- Research is an apolitical activity — no political bias

Errors in selecting a research problem

There could be many errors in selecting a research problem

- Population Specification — unknown whom or where to survey
- Sampling and Sample Frame Errors — wrong subpopulation is used to select a sample
- Selection — sample selected by a non-probability method
- Non-responsive — absence of data rather than inaccurate data
- Measurement — measurement error due to uncontrollable factors

Scope and objectives of research problem

Scopes of a research problem

- Topic coverage — Specify the topics, issues, or phenomena
- Target population — Define the group or demographic the research will study
- Geographic location — Identify the geographical area to conduct the research
- Time frame — time period for the data collection
- Variables or Factors — key variables, parameters, or factors the study
- Exclusion — where not to conduct research

How to write scopes of a research problem

Writing the scopes of a research problem

- Define Your Research Objectives — primary goals and objectives of your study
- Identify Boundaries — determine the specific aspects and exclude unrelated elements
- Justify Your Choices — explain logically why it is relevant
- Write a Clear Statement — summarize the scope

Reference to history

The history of world civilization speaks about a lot of information of the world.

These are living documents of the history of world research.

- Harappa and Mahenjodaro – what do these ancient civilization teach?
- The Greek civilization – what do the philosophers of this period teach?
- European industrial revolution – what scientific revolution did they achieve and apply?
- The Indian renaissance of the nineteenth century – how did the weavers of Bengal master the science and technology of weaving the finest linen called "Dhakai Maslin"?
- What about the wondrous scientific research and innovations by Jagadish Chandra Bose – (i) Wireless communication system discovery and development, and (ii) Developing the apparatus and method to show the breathing of the plants.

Reference to literature and art

- Rabindranath Thakur (Tagore) wrote a short book named, "Bishwa Parichay," describing astronomical research in simple Bangla.
- How could Rabindranath write those topics so correctly at a time when there were no Internet, or fax. Most of the astronomical research were published in German!
- Bibhuti Bhushan Bandyopadhyay wrote a novel called, "Chander Pahar," describing the story set in the dense Jungle of Africa. How could he write so much details without visiting and seeing Africa?
- In more recent days, Dinesh Chandra Chattopadhyay, the owner of the publisher Dev Sahitya Kutir, wrote a book entitled, "Duronto Eagle" set in Uzbekistan (erstwhile USSR). (Refer to an interview of Tridib Chattopadhyay in Mirchi Bangla about "Duronto Eagle.")
- Many of the literary works included deep research about places and lives of global citizens.

Reference to scientific discoveries and inventions

- How did the scientic discoveries happen?
- How the scientific innovations took shape in the past?
- We will discuss this topic in a future class.